Team Description Paper (TDP)

Team name: RoboKit

Affiliation: Technopark of the P. L. Kapitsa Phystech Lyceum

Country: Russia

City: Dolgoprudny

Team leader: Dmitry Dmitrievich Ustinov

Robot platform: Roki-2 (humanoid robot with 23 servomotors)

Website: https://starkit.su/

Abstract

MechTech RoboKit is a school robotics team based at the humanoid robotics laboratory of the Phystech Lyceum Technopark. Since 2019, we have been assembling, programming and controlling robots using computer vision and modern artificial intelligence tools.

We use Roki-2, a Russian humanoid platform developed at the Moscow Institute of Physics and Technology (MIPT). Students tune motions, develop behaviour algorithms and prepare robots for autonomous soccer and all-round events. Azer Babaev and MIPT's StarKit team support our work.

Competition Participation

RoboCup Asia-Pacific 2019 — 1st and 3rd places; Virtual RoboCup Asia-Pacific 2020 — 1st place. Robofest 2020 — 1st and 2nd in Junior Humanoid Soccer. RoboCup Russia 2021 — victory for Robokit A; FIRA 2021 — 2nd in the overall all-round standings.

RoboScience Hackathon 2023 — 12 podium finishes, including a clean sweep of the triple jump podium.

FIRA RoboWorld Cup and Summit 2024, Brazil — 1st and 2nd in Humanoid Junior. FIRA Korea 2024 — wins in HuroCup All Round, Kids Weightlifting and Marathon. FIRA Eurasia 2024, Türkiye — 1st places in the junior and adult categories.

RoboCup Russia 2024 — 1st and 2nd in autonomous soccer and 2nd in Entry. RoboScience Hackathon 2024 — 21 podium finishes in humanoid leagues: nine 1st, seven 2nd and five 3rd places.

Competition Participation (continued)

RoboCup Russia 2025 — nine podium finishes: three 1st, three 2nd and three 3rd places. The team swept the podium in autonomous soccer and triple jump, also taking 1st and 3rd in Entry and 2nd in sprint.

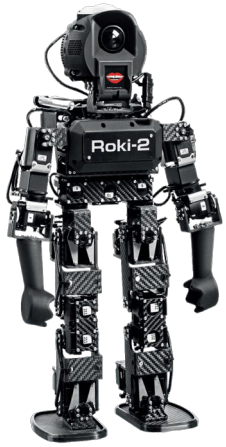
FIRA 2025, Taiwan — a clean sweep of the school league podium. FIRA RoboWorld Cup 2025, South Korea — 1st in penalty kicks and three 3rd places: triple jump, sprint and penalty kicks.

RoboScience Hackathon 2025 — a clean sweep in humanoid soccer, plus 1st and 2nd in junior all-round. At the Robofest Olympiad, Evgeny Kravets was the overall winner among ninth-grade students; three other students received third-degree diplomas.

RoboCup Russia 2026 — six podium finishes: 1st and 2nd in Junior Humanoid Soccer, 3rd in Entry, 2nd in marathon, 3rd in overall HuroCup and 3rd in sprint. At FIRA RoboSports Cup Asia 2026 in Taiwan, Fyodor Devyatov and Ilya Borisov won in the individual HuroCup standings.

Hardware Overview

Our main platform is Roki-2, the “Robot Athlete” kit developed at MIPT. It introduces students to humanoid assembly, actuator tuning and preparation for sports tasks. In 2025, we also competed with Roki-4 and the Khodok (“Walker”) platform.



The Roki-2 kit includes:

23 servomotors and a preassembled head module

Aluminium and carbon-fibre body parts, together with 3D-printed plastic components

Electronic boards, connecting cables and a battery-powered supply system

Charging equipment, a stand and a transport case

The robot can walk, jump, bend and get up after falling. Basic capabilities include finding coloured objects, line following and soccer. The development environment runs on an external Windows PC.

Roki-2 platform. Photo: STARKIT, sk24.ru

Software and AI

The platform provides predefined motions and control tools. We extend these with our own behaviour algorithms, image processing and motion settings tailored to competition tasks.

Our main areas of work include:

Python behaviour trees for selecting robot actions

OpenCV-based computer vision for object detection and image processing

Motion tuning and custom controllers for soccer, running, jumping and strength tasks

Training neural networks for object recognition, including classroom work with YOLOv8

In autonomous soccer, we combine ball search, approach and kicking. For marathon events, we practise line following; for all-round events, we tune motions for individual exercises. Students learn to integrate vision, decision-making and actuator control into a single program.

We use Elsiros/Webots simulation for learning and preparation. Neural locomotion control and reinforcement learning are research directions for future development.

Educational Mission

Working with Roki-2 gives students practical experience in mechanical and electrical assembly, Python programming, computer vision and neural networks. They progress from individual algorithms to a working robot and test their solutions in competitions.

Our educational programmes are free of charge. The laboratory holds weekly classes, lectures and practical sessions. We develop teamwork: students discuss solutions, share responsibilities, conduct experiments and pass on their experience to new members.

Our Goal

We aim to give students substantial engineering experience while maintaining strong performances in Russian and international competitions. Our next priorities are to improve the robustness of autonomous behaviour, enhance recovery after falls and expand algorithm testing in simulation and on real robots.